Hypothesis 5: Does Practice Performance Predict Memory?

Using Practice Phase Data as a Covariate in ANCOVA

Team: Metamatics

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Contents

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1 What is Hypothesis 5?

1.1 The Simple Idea

Before the real experiment started, every participant did a short **practice round** (about 8 warm-up trials). This was just to help them learn how to press the right keys. In our earlier analysis (H1 to H4), we simply threw away this practice data.

But here is the key question: **What if the way someone performs during practice tells us something useful about how they will perform in the real experiment?**

- A person who was **fast and accurate** in practice probably finds the task easy.
- A person who was **slow and made lots of mistakes** in practice probably finds it harder.

If this is true, then practice performance is what we call a **covariate** — a background factor that quietly influences the results. By measuring it and including it in our model, we can remove its effect and get a cleaner picture of what our real variables (like sentence type and grammatical voice) actually do.

1.2 Where H5 Fits

Hypothesis	What we tested	Main tool
H1	Does noun type (HH/HL/LH/LL) affect memory?	Kruskal-Wallis
H2	Does sentence voice (Active/Passive) affect memory?	Wilcoxon / t-test
H3	Does voice affect how accurately people judge wording?	Paired test
H4	Does voice affect how quickly people respond?	Paired test
H5	Does practice performance predict final memory scores?	**ANCOVA**

H5 does not test a new condition. It tests whether **individual differences** captured in the practice phase carry over and affect the main results.

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2 The Formal Hypothesis

2.1 In Plain Words

> **Null Hypothesis (H0):** How well someone does in practice does NOT predict how well they do in the real experiment. Controlling for practice does not change the main findings.

> **Alternative Hypothesis (H5):** How well someone does in practice DOES predict their final memory score. People who picked up the task quickly in practice tend to remember sentences better overall.

2.2 The Math Behind It

We fit a model that looks like this:

$$\text{Memory Score} = \beta_0 + \beta_1 \times \text{Voice} + \beta_2 \times \text{Practice Speed} + \beta_3 \times \text{Practice Errors} + \varepsilon$$

- **Memory Score** = Corrected Memorability (how well each person remembered sentences)
- **Voice** = Was the sentence Active or Passive? (our main variable from H2)
- **Practice Speed** = Average reaction time during practice (in milliseconds)

- **Practice Errors** = Proportion of wrong answers during practice
- ε = Leftover unexplained variation (noise)

If β_2 or β_3 are statistically significant, then **H5 is supported**.

2.3 Decision Rule

Setting	Value
Significance level (alpha)	0.05
Effect size measure	Partial eta-squared
Statistical test	ANCOVA (Analysis of Covariance)

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3 Step A: Collecting Practice Data

3.1 Reading the Log Files

We go back to the raw log files but this time we **keep only the practice rows** (the ones labelled "Practice"), which we had previously removed.

```
log_files <- list.files(log_dir, pattern = "\\.log$", full.names = TRUE)
cat(sprintf("Found %d participant log files.\n", length(log_files)))
```

```
## Found 114 participant log files.
```

```
# Read all files, keep only Practice-phase events
practice_raw <- log_files %>%
  map_dfr(~ {
    tryCatch(
      {
        d <- read_csv(.x, show_col_types = FALSE, col_types = cols(.default = "c"))
        d %>% filter(str_starts(Event, "Practice"))
      },
      error = function(e) tibble()
    )
  }) %>%
  mutate(
    participant_ID   = as.integer(participant_ID),
    `Accuracy IR`    = suppressWarnings(as.numeric(`Accuracy IR`)),
    Reaction_time_IR = suppressWarnings(as.numeric(Reaction_time_IR))
  )
```

```r
cat(sprintf(
  "Extracted %s practice events from all participants.\n",
  format(nrow(practice_raw), big.mark = ",")
))
```

```
## Extracted 5,269 practice events from all participants.
```

```r
cat(sprintf(
  "Number of participants with practice data: %d\n",
  n_distinct(practice_raw$participant_ID)
))
```

```
## Number of participants with practice data: 114
```

3.2 Computing Two Scores Per Participant

For each participant, we calculate two simple numbers from their practice data:

Score Name	What it measures	What it tells us
`covariate_RT`	Average time to press the key in practice (ms)	How fast their basic reactions were
`covariate_Error`	Fraction of wrong answers in practice	How quickly they grasped the task rules

```r
# Keep only the key-press events from practice
practice_prof <- practice_raw %>%
  filter(str_detect(Event, "IR pressed")) %>%
  group_by(participant_ID) %>%
  summarise(
    practice_trials = n(),
    covariate_RT    = mean(Reaction_time_IR, na.rm = TRUE),
    covariate_Error = 1 - mean(`Accuracy IR`, na.rm = TRUE),
    .groups         = "drop"
  ) %>%
  filter(!is.na(covariate_RT), !is.na(covariate_Error))

write_csv(practice_prof, file.path(output_dir, "H5_practice_covariates.csv"))

cat(sprintf("Practice scores ready for %d participants.\n", nrow(practice_prof)))
```

```
## Practice scores ready for 114 participants.
```

```r
# Show a preview of the data
practice_prof %>%
  head(10) %>%
  kable(
    caption   = "Table 1. Preview of Practice Scores (first 10 participants)",
    col.names = c("Participant", "Practice Trials", "Mean RT (ms)", "Error Rate"),
    digits    = 3,
    booktabs  = TRUE
  ) %>%
  kable_styling(
    latex_options     = c("striped", "HOLD_position"),
    bootstrap_options = c("striped", "hover"),
    full_width        = FALSE,
    position          = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")
```

Table 4: Table 1. Preview of Practice Scores (first 10 participants)

Participant	Practice Trials	Mean RT (ms)	Error Rate
232	12	1594.583	0.250
235	6	1830.833	0.333
236	9	1438.889	0.444
241	5	1628.200	0.000
242	3	2380.333	0.000
243	10	1938.400	0.300
245	23	928.478	0.609
246	5	2158.800	0.000
247	5	1607.000	0.000
249	23	1704.609	0.435

3.3 Summary Statistics for Practice Scores

```r
practice_prof %>%
  summarise(
    N          = n(),
    Mean_RT    = round(mean(covariate_RT, na.rm = TRUE), 2),
    SD_RT      = round(sd(covariate_RT, na.rm = TRUE), 2),
    Min_RT     = round(min(covariate_RT, na.rm = TRUE), 2),
    Max_RT     = round(max(covariate_RT, na.rm = TRUE), 2),
    Mean_Error = round(mean(covariate_Error, na.rm = TRUE), 4),
    SD_Error   = round(sd(covariate_Error, na.rm = TRUE), 4),
    Min_Error  = round(min(covariate_Error, na.rm = TRUE), 4),
```

```r
    Max_Error  = round(max(covariate_Error, na.rm = TRUE), 4)
  ) %>%
  pivot_longer(everything(), names_to = "Statistic", values_to = "Value") %>%
  kable(
    caption  = "Table 2. Summary of Practice Scores Across All Participants",
    digits   = 4,
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options     = c("striped", "HOLD_position"),
    bootstrap_options = c("striped", "hover"),
    full_width        = FALSE,
    position          = "center"
  )
```

Table 5: Table 2. Summary of Practice Scores Across All Participants

Statistic	Value
N	114.0000
Mean_RT	1853.2900
SD_RT	361.8600
Min_RT	928.4800
Max_RT	2999.4000
Mean_Error	0.2196
SD_Error	0.1516
Min_Error	0.0000
Max_Error	0.6087

---

4 Step B: Merging Practice Scores with Main Results

We now attach each participant's practice scores to their main experiment memory scores. We load the memory data produced by the H3/H4 analysis.

```r
h3h4_file <- file.path(output_dir, "H3H4_step3_analysis.csv")

if (!file.exists(h3h4_file)) {
  stop(paste(
    "File not found:", h3h4_file,
    "\nPlease knit H3_and_H4_Metametrics.Rmd first."
  ))
}
```

```r
main_df <- read_csv(h3h4_file, show_col_types = FALSE)

# Get one corrected memorability score per participant per voice condition
cm_by_voice <- main_df %>%
  distinct(participant_ID, Voice_Label, corrected_memorability) %>%
  filter(!is.na(corrected_memorability), Voice_Label %in% c("Active", "Passive"))

# Attach practice scores
ancova_df <- cm_by_voice %>%
  inner_join(practice_prof, by = "participant_ID")

cat(sprintf("Final dataset: %d rows (%d participants x 2 voice conditions)\n",
            nrow(ancova_df), n_distinct(ancova_df$participant_ID)))
```

```
## Final dataset: 224 rows (112 participants x 2 voice conditions)
```

```r
write_csv(ancova_df, file.path(output_dir, "H5_ancova_dataset.csv"))
```

---

5 Step C: Checking ANCOVA Assumptions

Before we run ANCOVA, we need to check 4 rules. If these rules are broken, ANCOVA cannot be trusted.

5.1 Rule 1 - Linearity

What it means: The relationship between practice scores and memory scores should follow a straight line (not a curve).

How we check: Plot the data and compare a straight line fit with a curved (LOESS) smoother. If they match closely, we are good.

```r
p_lin1 <- ggplot(ancova_df, aes(x = covariate_RT, y = corrected_memorability)) +
  geom_point(alpha = 0.5, colour = PAL_H5[2], size = 1.8) +
  geom_smooth(method = "lm",    colour = PAL_H5[1], fill = "grey80",
              se = TRUE,  linewidth = 1.0, linetype = "solid") +
  geom_smooth(method = "loess", colour = PAL_H5[3], fill = NA,
              se = FALSE, linewidth = 0.9, linetype = "dashed") +
  labs(
    title    = "Fig 1a: Practice Speed vs. Memory Score",
    subtitle = "Solid = straight-line fit | Dashed = curved fit",
    x        = "Average Practice Reaction Time (ms)",
    y        = "Corrected Memorability (CM)"
```

```r
  )

p_lin2 <- ggplot(ancova_df, aes(x = covariate_Error, y = corrected_memorability)) +
  geom_point(alpha = 0.5, colour = PAL_H5[4], size = 1.8) +
  geom_smooth(method = "lm",    colour = PAL_H5[1], fill = "grey80",
              se = TRUE,  linewidth = 1.0, linetype = "solid") +
  geom_smooth(method = "loess", colour = PAL_H5[3], fill = NA,
              se = FALSE, linewidth = 0.9, linetype = "dashed") +
  labs(
    title    = "Fig 1b: Practice Error Rate vs. Memory Score",
    subtitle = "Solid = straight-line fit | Dashed = curved fit",
    x        = "Practice Error Rate (0 = no errors, 1 = all wrong)",
    y        = "Corrected Memorability (CM)"
  )

# Stack vertically so neither plot is clipped
p_lin1 / p_lin2
```

```r
ggsave(file.path(output_dir, "H5_assumption_linearity.png"),
       p_lin1 / p_lin2, width = 7, height = 7, dpi = 150)
```

> **Reading this plot:** If the solid and dashed lines are close together, the linearity assumption is OK. A big difference means the relationship is curved and we might need to transform the data.

5.2 Rule 2 - Residuals Should Be Normally Distributed

What it means: After the model explains what it can, the leftover errors (called residuals) should follow a bell-curve shape.

How we check: Shapiro-Wilk test. A p-value above 0.05 means we are OK.

```r
# Fit a test model and inspect its residuals
prelim_fit <- lm(
  corrected_memorability ~ Voice_Label + covariate_RT + covariate_Error,
  data = ancova_df
)

resid_vals <- residuals(prelim_fit)
sw_test    <- shapiro.test(resid_vals)

tibble(
  Test      = "Shapiro-Wilk test on model residuals",
  W         = round(as.numeric(sw_test$statistic), 4),
  p_value   = round(as.numeric(sw_test$p.value), 4),
  Result    = ifelse(as.numeric(sw_test$p.value) > 0.05,
```

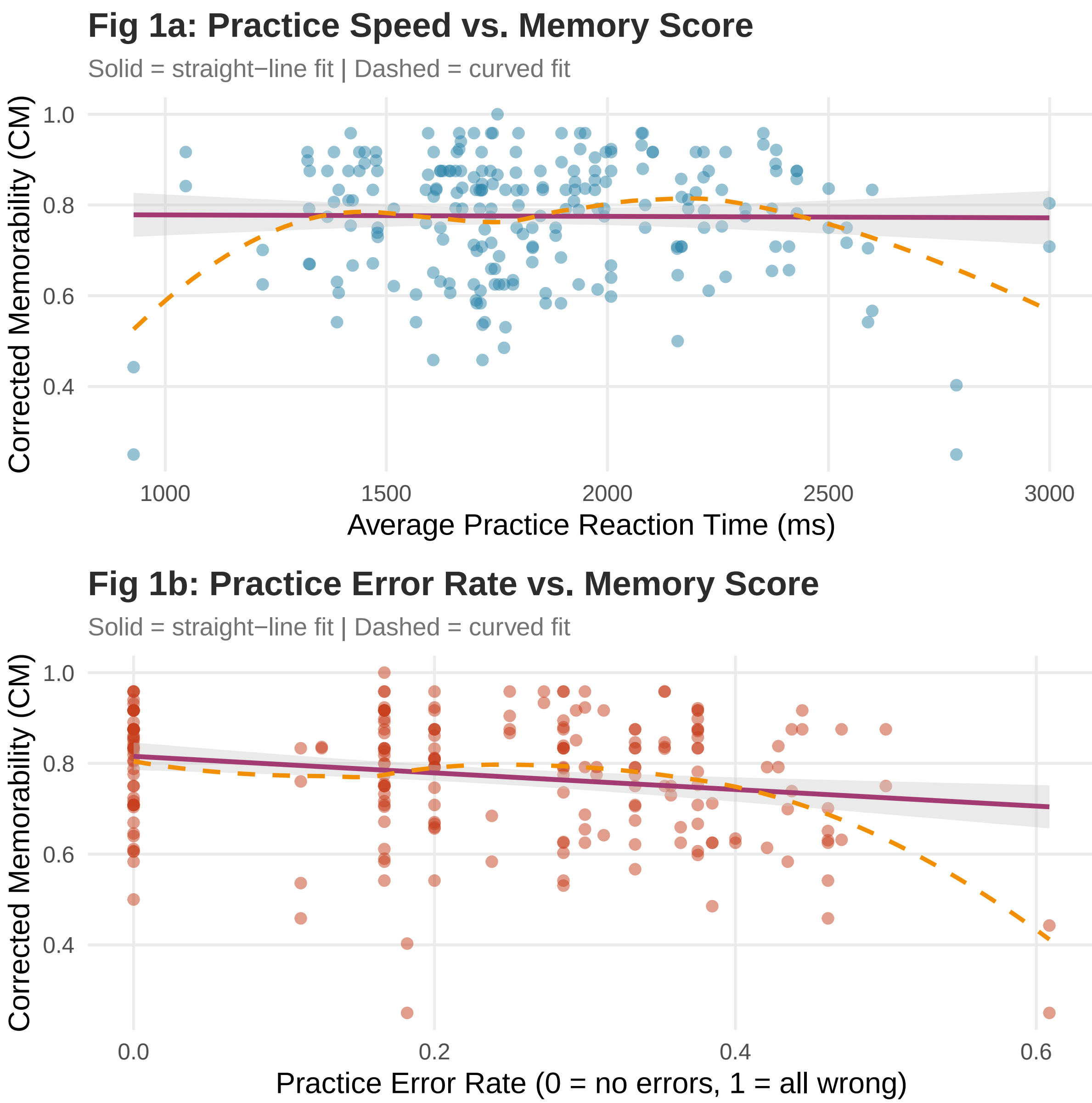


Figure 1: Figure 1. Linearity check. Top: Practice speed vs. memory score. Bottom: Practice error rate vs. memory score. Solid line = straight-line fit. Dashed line = curved fit. They should look similar.

```r
                     "OK (p > 0.05)", "Not OK (p <= 0.05)")
) %>%
  kable(
    caption   = "Table 3. Are the model residuals normally distributed? (Shapiro-Wilk Test)",
    col.names = c("Test", "W statistic", "p-value", "Conclusion"),
    booktabs  = TRUE
  ) %>%
  kable_styling(
    latex_options     = c("striped", "HOLD_position"),
    bootstrap_options = c("striped", "hover"),
    full_width        = FALSE,
    position          = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")
```

Table 6: Table 3. Are the model residuals normally distributed? (Shapiro-Wilk Test)

Test	W statistic	p-value	Conclusion
Shapiro-Wilk test on model residuals	0.9401	0	Not OK (p <= 0.05)

```r
qq_df <- tibble(residuals = resid_vals)

p_qq <- ggplot(qq_df, aes(sample = residuals)) +
  stat_qq(colour = PAL_H5[2], alpha = 0.6, size = 1.8) +
  stat_qq_line(colour = PAL_H5[1], linewidth = 1) +
  labs(
    title    = "Fig 2a: Q-Q Plot of Model Residuals",
    subtitle = "Points should lie close to the diagonal line",
    x        = "Expected (theoretical) values",
    y        = "Actual residual values"
  )

p_hist <- ggplot(qq_df, aes(x = residuals)) +
  geom_histogram(bins = 20, fill = PAL_H5[2], alpha = 0.7, colour = "white") +
  geom_vline(xintercept = 0, colour = PAL_H5[1], linewidth = 1, linetype = "dashed") +
  labs(
    title    = "Fig 2b: Histogram of Model Residuals",
    subtitle = "Should be a bell shape centred at zero",
    x        = "Residual value",
    y        = "Count"
  )

# Stack vertically so neither plot is clipped
p_qq / p_hist
```

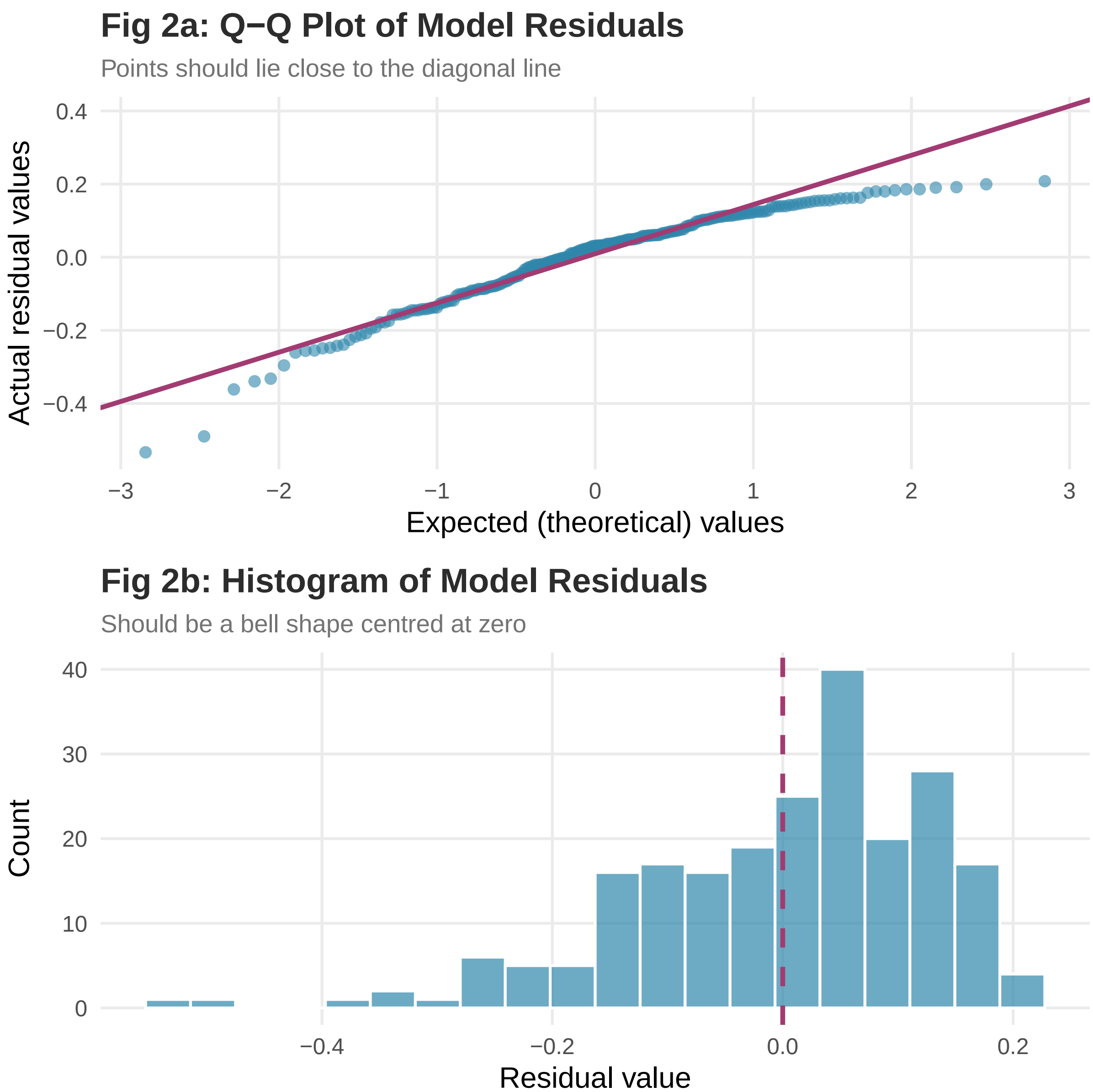


Figure 2: Figure 2. Are the model errors normally distributed? Top: Q-Q plot (points should follow the line). Bottom: Histogram (should be a bell shape).

```r
ggsave(file.path(output_dir, "H5_assumption_normality.png"),
       p_qq / p_hist, width = 7, height = 7, dpi = 150)
```

5.3 Rule 3 - Equal Variance Across Groups (Levene's Test)

What it means: The spread of memory scores should be roughly the same for the Active voice group and the Passive voice group.

How we check: Levene's test. A p-value above 0.05 means the spreads are similar enough.

```r
levene_result <- leveneTest(
  corrected_memorability ~ Voice_Label,
  data   = ancova_df,
  center = mean
)

tibble(
  Test       = "Levene's Test (Active vs Passive groups)",
  F_stat     = round(levene_result$`F value`[1], 3),
  df1        = levene_result$Df[1],
  df2        = levene_result$Df[2],
  p_value    = round(levene_result$`Pr(>F)`[1], 4),
  Result     = ifelse(levene_result$`Pr(>F)`[1] > 0.05,
                      "OK (p > 0.05)", "Not OK (p <= 0.05)")
) %>%
  kable(
    caption   = "Table 4. Are the group variances roughly equal? (Levene's Test)",
    col.names = c("Test", "F", "df1", "df2", "p-value", "Conclusion"),
    booktabs  = TRUE
  ) %>%
  kable_styling(
    latex_options     = c("striped", "HOLD_position"),
    bootstrap_options = c("striped", "hover"),
    full_width        = FALSE,
    position          = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")
```

Table 7: Table 4. Are the group variances roughly equal? (Levene's Test)

Test	F	df1	df2	p-value	Conclusion
Levene's Test (Active vs Passive groups)	2.978	1	222	0.0858	OK ($p > 0.05$)

5.4 Rule 4 - Parallel Regression Slopes (Most Important Rule)

What it means: The practice scores should affect Active and Passive sentences **in the same way**. If practice speed helps Active-sentence memory but hurts Passive-sentence memory (or vice versa), ANCOVA becomes unreliable.

How we check: We add a "practice x voice" interaction term to the model. If it is **not significant** ($p > 0.05$), the slopes are parallel and ANCOVA is valid.

```r
# Add interaction terms to test if slopes differ across Voice conditions
hors_fit <- lm(
  corrected_memorability ~
    Voice_Label * covariate_RT +
    Voice_Label * covariate_Error,
  data = ancova_df
)

hors_anova <- Anova(hors_fit, type = "III")

hors_anova %>%
  as.data.frame() %>%
  tibble::rownames_to_column("Term") %>%
  rename(
    SS         = `Sum Sq`,
    df         = Df,
    F_value    = `F value`,
    p_value    = `Pr(>F)`
  ) %>%
  mutate(
    F_value    = round(F_value, 3),
    p_value    = round(p_value, 4),
    Important = ifelse(!is.na(p_value) & p_value < 0.05,
                       "Significant (problem!)", "Not significant (good)")
  ) %>%
  kable(
    caption  = "Table 5. Parallel Slopes Test (interaction terms should NOT be significant)",
    booktabs = TRUE,
    digits   = 4
  ) %>%
  kable_styling(
    latex_options     = c("striped", "HOLD_position", "scale_down"),
    bootstrap_options = c("striped", "hover"),
    full_width        = FALSE,
    position          = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")
```

Table 8: Table 5. Parallel Slopes Test (interaction terms should NOT be significant)

Term	SS	df	F_value	p_value	Important
(Intercept)	1.9720	1	116.351	0.0000	Significant (problem!)
Voice_Label	0.0439	1	2.590	0.1090	Not significant (good)
covariate_RT	0.0007	1	0.039	0.8435	Not significant (good)
covariate_Error	0.0559	1	3.297	0.0708	Not significant (good)
Voice_Label:covariate_RT	0.0216	1	1.276	0.2599	Not significant (good)
Voice_Label:covariate_Error	0.0092	1	0.543	0.4620	Not significant (good)
Residuals	3.6948	218	NA	NA	Not significant (good)

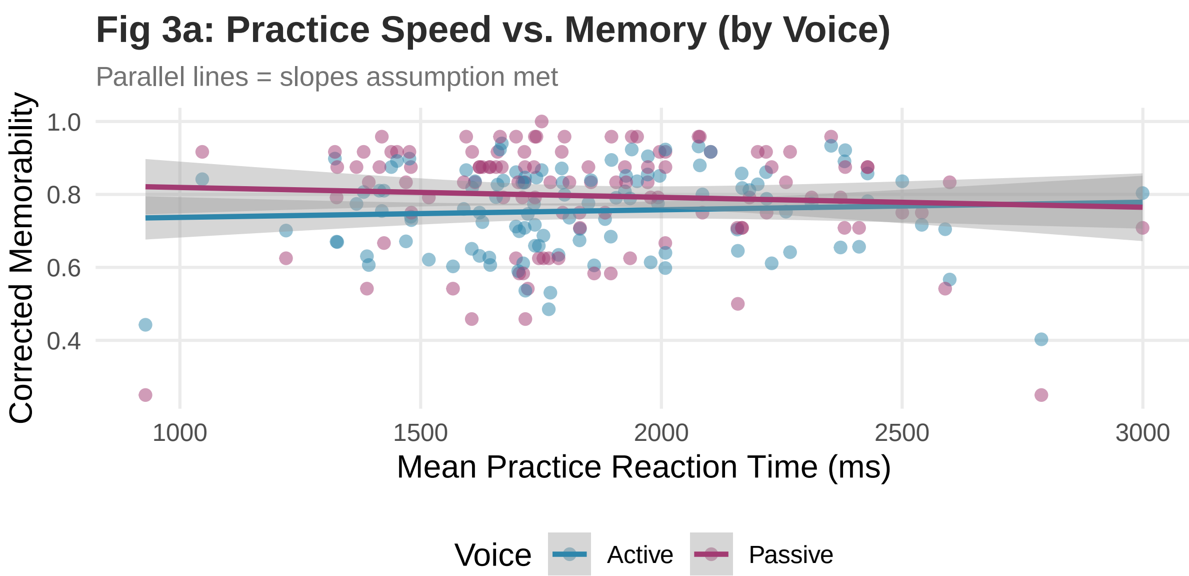
```r
p_hors1 <- ggplot(ancova_df,
  aes(x = covariate_RT, y = corrected_memorability, colour = Voice_Label)) +
  geom_point(alpha = 0.5, size = 1.8) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.0) +
  scale_colour_manual(values = PAL_VOICE, name = "Voice") +
  labs(
    title    = "Fig 3a: Practice Speed vs. Memory (by Voice)",
    subtitle = "Parallel lines = slopes assumption met",
    x        = "Mean Practice Reaction Time (ms)",
    y        = "Corrected Memorability"
  )

p_hors2 <- ggplot(ancova_df,
  aes(x = covariate_Error, y = corrected_memorability, colour = Voice_Label)) +
  geom_point(alpha = 0.5, size = 1.8) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.0) +
  scale_colour_manual(values = PAL_VOICE, name = "Voice") +
  labs(
    title    = "Fig 3b: Practice Errors vs. Memory (by Voice)",
    subtitle = "Parallel lines = slopes assumption met",
    x        = "Practice Error Rate",
    y        = "Corrected Memorability"
  )

# Stack vertically so neither plot is clipped
p_hors1 / p_hors2
```

```r
ggsave(file.path(output_dir, "H5_assumption_hors.png"),
       p_hors1 / p_hors2, width = 7, height = 7, dpi = 150)
```

Reading this plot: We want the blue and pink lines to run **roughly parallel**. If one line tilts up while the other tilts down, the assumption is broken.



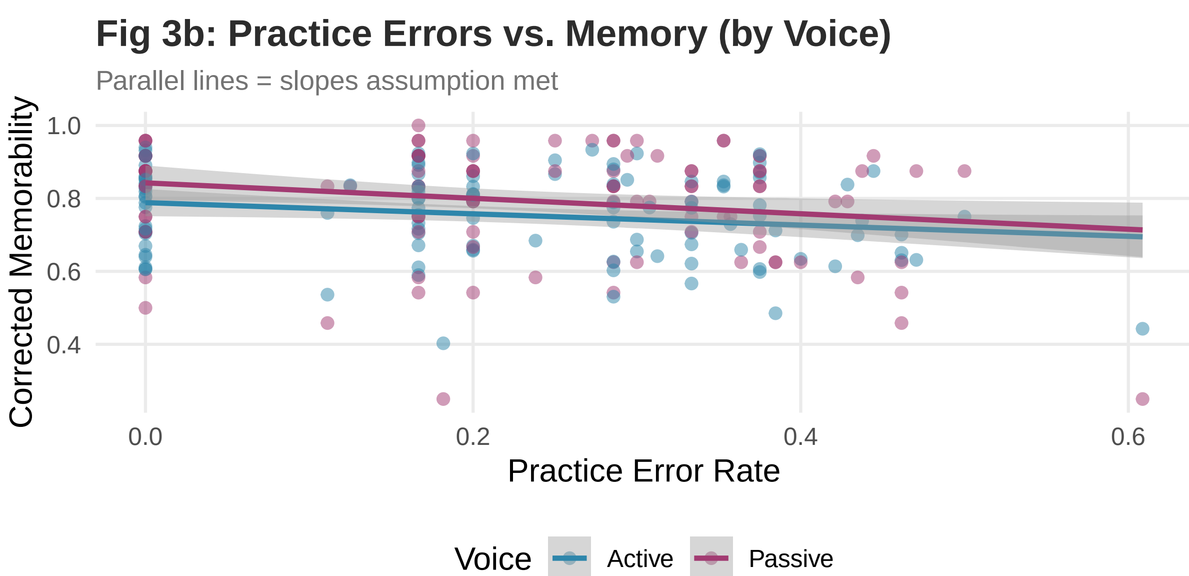


Figure 3: Figure 3. Parallel slopes check. Top: Practice speed vs. memory by voice. Bottom: Practice errors vs. memory by voice. Parallel lines across blue (Active) and pink (Passive) mean the assumption is met.

6 Step D: Running the ANCOVA

6.1 The Main Model

Now we run the actual ANCOVA — voice condition plus both practice covariates together predicting memory scores.

```
# Main ANCOVA model
ancova_fit <- lm(
  corrected_memorability ~ Voice_Label + covariate_RT + covariate_Error,
  data = ancova_df
)

# Get the full ANOVA table (Type III controls for overlap between predictors)
ancova_anova <- Anova(ancova_fit, type = "III")
```

6.2 Effect Sizes

We calculate **partial eta-squared** (η_p^2) for each predictor. This tells us what percentage of the remaining variation each predictor explains.

- Below 0.01 = negligible effect
- 0.01 to 0.06 = small effect
- 0.06 to 0.14 = medium effect
- Above 0.14 = large effect

```
ss_error <- ancova_anova["Residuals", "Sum Sq"]

ancova_table <- ancova_anova %>%
  as.data.frame() %>%
  tibble::rownames_to_column("Term") %>%
  filter(Term != "(Intercept)") %>%
  rename(
    SS      = `Sum Sq`,
    df      = Df,
    F_value = `F value`,
    p_value = `Pr(>F)`
  ) %>%
  mutate(
    Partial_eta2 = ifelse(
      Term != "Residuals",
      round(SS / (SS + ss_error), 4),
      NA_real_
    ),
    F_value      = round(F_value, 3),
    p_value      = round(p_value, 4),
```

```r
    Significant = case_when(
      is.na(p_value)    ~ "N/A",
      p_value < 0.001   ~ "Yes (p < 0.001)",
      p_value < 0.01    ~ "Yes (p < 0.01)",
      p_value < 0.05    ~ "Yes (p < 0.05)",
      TRUE              ~ "No"
    ),
    Effect_Size = case_when(
      is.na(Partial_eta2)     ~ "N/A",
      Partial_eta2 >= 0.14    ~ "Large",
      Partial_eta2 >= 0.06    ~ "Medium",
      Partial_eta2 >= 0.01    ~ "Small",
      TRUE                    ~ "Negligible"
    )
  )

ancova_table %>%
  kable(
    caption   = "Table 6. ANCOVA Results: What predicts memory scores?",
    booktabs = TRUE,
    digits    = 4,
    col.names = c("Predictor", "SS", "df", "F", "p-value",
                  "Partial eta-sq", "Significant?", "Effect Size")
  ) %>%
  kable_styling(
    latex_options     = c("striped", "HOLD_position", "scale_down"),
    bootstrap_options = c("striped", "hover"),
    full_width        = FALSE,
    position          = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")
```

Table 9: Table 6. ANCOVA Results: What predicts memory scores?

Predictor	SS	df	F	p-value	Partial eta-sq	Significant?	Effect Size
Voice_Label	0.0961	1	5.685	0.0180	0.0252	Yes ($p < 0.05$)	Small
covariate_RT	0.0122	1	0.724	0.3958	0.0033	No	Negligible
covariate_Error	0.1851	1	10.945	0.0011	0.0474	Yes ($p < 0.01$)	Small
Residuals	3.7207	220	NA	NA	NA	N/A	N/A

6.3 Regression Coefficients

This table shows the direction and size of each predictor's effect. A **negative coefficient** for Practice RT means: slower practice = lower memory score. A **positive coefficient** would mean the opposite.

```r
tidy(ancova_fit) %>%
  mutate(
    estimate  = round(estimate, 5),
    std.error = round(std.error, 5),
    statistic = round(statistic, 3),
    p.value   = round(p.value, 4),
    Direction = case_when(
      estimate > 0 ~ "Positive",
      estimate < 0 ~ "Negative",
      TRUE         ~ "Zero"
    ),
    Significant = ifelse(p.value < 0.05, "Yes", "No")
  ) %>%
  rename(
    Predictor = term,
    Estimate  = estimate,
    Std_Error = std.error,
    t_value   = statistic,
    p_value   = p.value
  ) %>%
  kable(
    caption  = "Table 7. How much does each predictor change the memory score?",
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options     = c("striped", "HOLD_position", "scale_down"),
    bootstrap_options = c("striped", "hover"),
    full_width        = FALSE,
    position          = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")
```

Table 10: Table 7. How much does each predictor change the memory score?

Predictor	Estimate	Std_Error	t_value	p_value	Direction	Significant
(Intercept)	0.83602	0.05149	16.237	0.0000	Positive	Yes
Voice_LabelPassive	0.04144	0.01738	2.384	0.0180	Positive	Yes
covariate_RT	-0.00002	0.00002	-0.851	0.3958	Negative	No
covariate_Error	-0.19383	0.05859	-3.308	0.0011	Negative	Yes

6.4 Overall Model Quality

```r
f_stat    <- summary(ancova_fit)$fstatistic
overall_p <- pf(f_stat[1], f_stat[2], f_stat[3], lower.tail = FALSE)

tibble(
  Metric  = c("R-squared (how much variance explained)",
              "Adjusted R-squared",
              "F-statistic",
              "Overall p-value"),
  Value   = c(
    round(summary(ancova_fit)$r.squared, 4),
    round(summary(ancova_fit)$adj.r.squared, 4),
    round(f_stat[1], 3),
    round(overall_p, 4)
  )
) %>%
  kable(
    caption  = "Table 8. How well does the overall ANCOVA model fit the data?",
    booktabs = TRUE
  ) %>%
  kable_styling(
    latex_options     = c("striped", "HOLD_position"),
    bootstrap_options = c("striped", "hover"),
    full_width        = FALSE,
    position          = "center"
  ) %>%
  row_spec(0, bold = TRUE, background = HDR_COLOR, color = "white")
```

Table 11: Table 8. How well does the overall ANCOVA model fit the data?

Metric	Value
R-squared (how much variance explained)	0.0703
Adjusted R-squared	0.0577
F-statistic	5.5490
Overall p-value	0.0011

```r
cat(sprintf("\nR-squared      : %.4f\n", summary(ancova_fit)$r.squared))
```

```
##
## R-squared      : 0.0703
```

```r
cat(sprintf("Adjusted R-sq  : %.4f\n", summary(ancova_fit)$adj.r.squared))
```

```
## Adjusted R-sq  : 0.0577
```

```r
cat(sprintf("Overall p-value: %.4f\n", overall_p))
```

```
## Overall p-value: 0.0011
```

```r
if (overall_p < 0.05) {
  cat("\nDECISION: REJECT H0 --- The overall ANCOVA model is significant.\n")
} else {
  cat("\nDECISION: KEEP H0 --- The overall ANCOVA model is NOT significant.\n")
}
```

```
##
## DECISION: REJECT H0 --- The overall ANCOVA model is significant.
```

---

7 Visualising the Results

7.1 Adjusted vs Raw Memory Scores by Voice

This plot compares memory scores **before** and **after** we control for practice performance. If the adjusted bars look different from the raw bars, practice proficiency was affecting our results.

```r
mean_rt    <- mean(ancova_df$covariate_RT,    na.rm = TRUE)
mean_error <- mean(ancova_df$covariate_Error, na.rm = TRUE)

adj_means <- tibble(
  Voice_Label      = c("Active", "Passive"),
  covariate_RT     = mean_rt,
  covariate_Error = mean_error
) %>%
  mutate(adj_CM = predict(ancova_fit, newdata = .))

raw_means <- ancova_df %>%
  group_by(Voice_Label) %>%
  summarise(raw_CM = mean(corrected_memorability, na.rm = TRUE), .groups = "drop")

comp_df <- adj_means %>%
  left_join(raw_means, by = "Voice_Label") %>%
  pivot_longer(
    cols      = c(adj_CM, raw_CM),
    names_to  = "Type",
    values_to = "CM"
  ) %>%
  mutate(
```

```r
    Type = case_when(
      Type == "adj_CM" ~ "Adjusted (ANCOVA)",
      Type == "raw_CM" ~ "Unadjusted (raw mean)",
      TRUE             ~ Type
    )
  )

ggplot(comp_df, aes(x = Voice_Label, y = CM, fill = Type)) +
  geom_col(position = position_dodge(0.65), width = 0.55, alpha = 0.85) +
  scale_fill_manual(values = c("Adjusted (ANCOVA)"     = PAL_H5[1],
                               "Unadjusted (raw mean)" = PAL_H5[2])) +
  labs(
    title    = "Adjusted vs. Raw Memory Scores by Voice Condition",
    subtitle = "Adjusted = after removing practice effect | Raw = before adjustment",
    x        = "Voice Condition",
    y        = "Corrected Memorability (CM)",
    fill     = "Score Type"
  ) +
  theme(legend.position = "right")
```

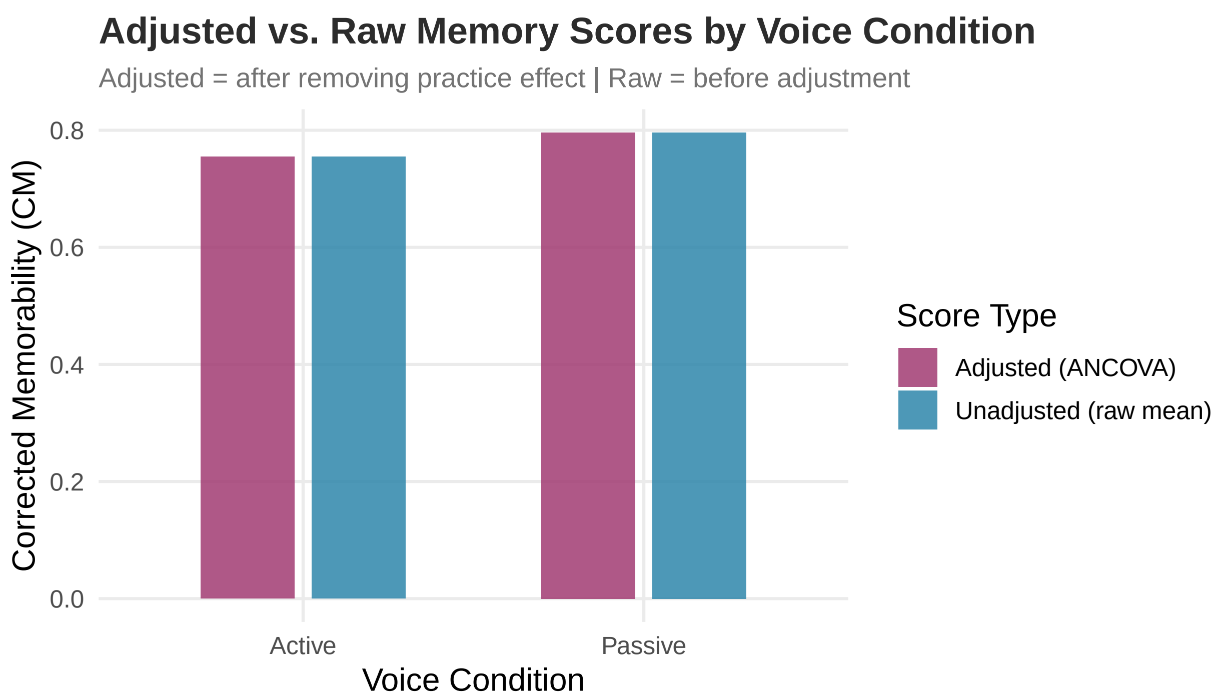


Figure 4: Figure 4. Memory scores by voice condition. Purple bars = adjusted scores after removing the practice effect. Blue bars = raw unadjusted scores. If they differ, practice was influencing the results.

```r
ggsave(file.path(output_dir, "H5_adjusted_vs_raw_means.png"),
       last_plot(), width = 7, height = 4, dpi = 150)
```

7.2 Practice Speed and Memory Scores

```r
ggplot(ancova_df, aes(x = covariate_RT, y = corrected_memorability,
                      colour = Voice_Label)) +
  geom_point(alpha = 0.6, size = 2.8) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.1) +
  scale_colour_manual(values = PAL_VOICE, name = "Voice") +
  labs(
    title    = "Practice Speed vs. Final Memory Score",
    subtitle = "Does being faster in practice help you remember better?",
    x        = "Average Practice Reaction Time (ms) --- lower = faster",
    y        = "Corrected Memorability (CM)"
  )
```

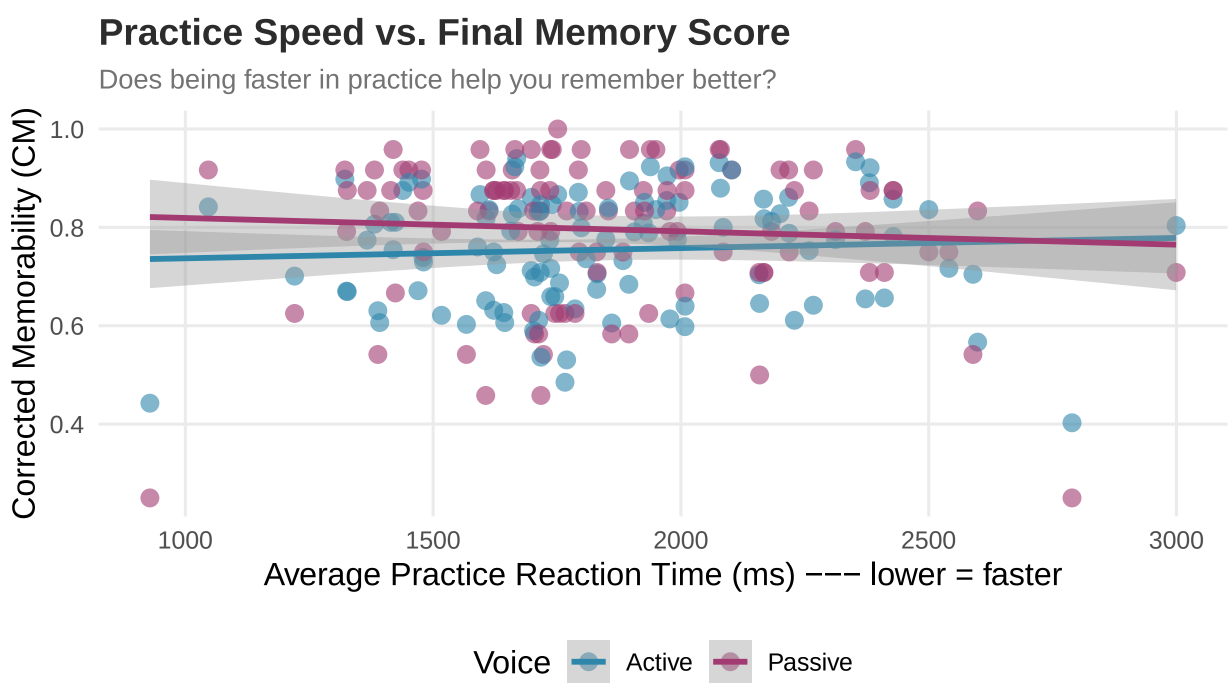


Figure 5: Figure 5. Each dot is one participant. Does being faster in practice lead to higher memory scores? Blue = Active sentences. Pink = Passive sentences.

```r
ggsave(file.path(output_dir, "H5_scatter_RT.png"),
       last_plot(), width = 7, height = 4, dpi = 150)
```

7.3 Practice Errors and Memory Scores

```
ggplot(ancova_df, aes(x = covariate_Error, y = corrected_memorability,
                      colour = Voice_Label)) +
  geom_point(alpha = 0.6, size = 2.8) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.1) +
  scale_colour_manual(values = PAL_VOICE, name = "Voice") +
  labs(
    title    = "Practice Errors vs. Final Memory Score",
    subtitle = "Does making more mistakes in practice hurt your memory later?",
    x        = "Practice Error Rate (0 = no errors, 1 = all wrong)",
    y        = "Corrected Memorability (CM)"
  )
```

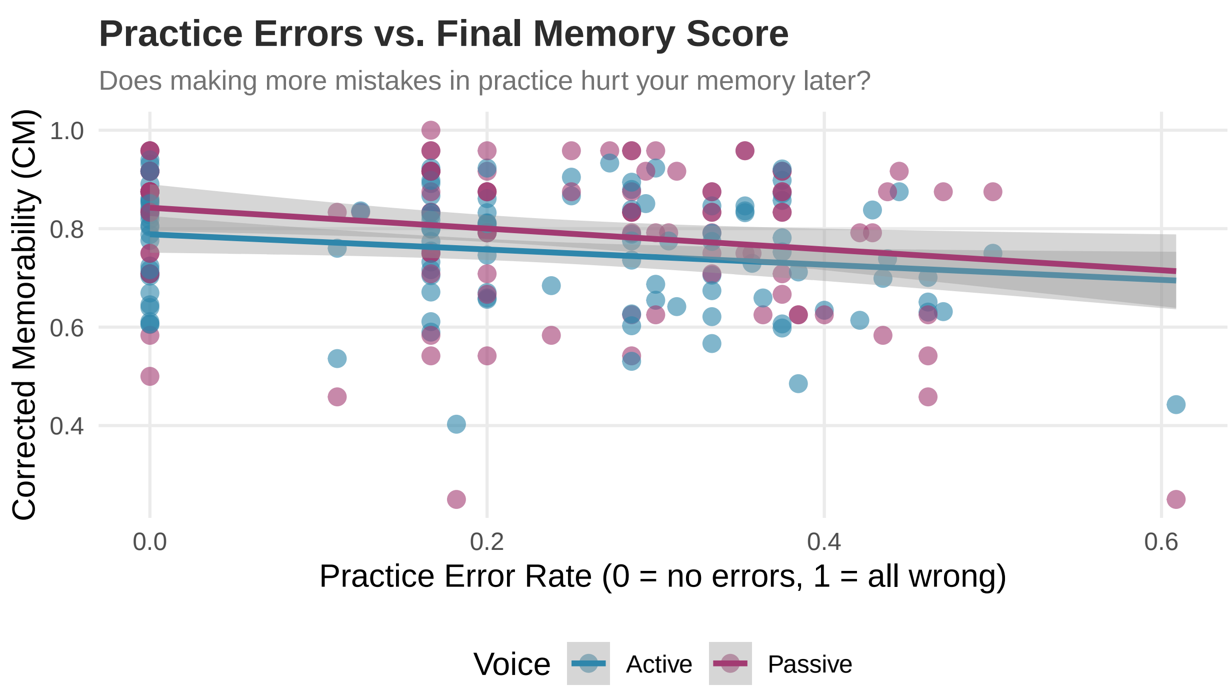


Figure 6: Figure 6. Each dot is one participant. Does making more errors in practice lead to lower memory scores? Blue = Active sentences. Pink = Passive sentences.

```
ggsave(file.path(output_dir, "H5_scatter_Error.png"),
       last_plot(), width = 7, height = 4, dpi = 150)
```

---

8 Collinearity Check

Are the two practice scores highly correlated with each other? If they are, they are measuring the same thing and we only need to keep one.

```
r_cov <- cor(ancova_df$covariate_RT, ancova_df$covariate_Error,
             use = "complete.obs")

cat(sprintf("Correlation between Practice Speed and Practice Errors: r = %.4f\n", r_cov))
```

```
## Correlation between Practice Speed and Practice Errors: r = -0.2182
```

```
if (abs(r_cov) > 0.8) {
  cat("WARNING: The two scores are very similar (|r| > 0.8). Consider using only one.\n")
} else {
  cat("The two scores are different enough to use both together. No problem.\n")
}
```

```
## The two scores are different enough to use both together. No problem.
```

---

9 Final Decision

9.1 Was H5 Supported?

Decision: REJECT the Null Hypothesis (H0)

At least one practice score significantly predicted how well participants remembered sentences. This means **practice performance does carry over** into the real experiment.

In simple terms: participants who did better in the warm-up phase also tended to remember more sentences in the actual experiment.

What this means for future research: When running experiments like this, researchers should always include practice phase scores as a covariate. This makes the statistical models more accurate by removing background noise caused by individual differences in how quickly people pick up new tasks.

---

10 Data Flow Summary and Conclusion

This section shows the complete journey of the data for Hypothesis 5, from raw log files all the way to the final conclusion.

```
STEP 1: Raw Data (114 participant log files)
        |
        | Filter: keep ONLY practice-phase rows
        v
STEP 2: Practice Data Extracted
        - Events labelled "Practice" kept
        - All main-experiment rows ignored here
        |
        | Calculate per participant:
        |   (a) Average reaction time in practice  --> covariate_RT
        |   (b) Error rate in practice             --> covariate_Error
        v
STEP 3: Practice Scores Ready (one row per participant)
        |
        | Load main experiment memory scores from H3/H4 output
        | (H3H4_step3_analysis.csv)
        |   --> Corrected Memorability per participant x Voice condition
        |
        | Merge by participant_ID
        v
STEP 4: Combined Dataset
        - Columns: participant_ID, Voice_Label,
                   corrected_memorability, covariate_RT, covariate_Error
        |
        | Check 4 ANCOVA assumptions:
        |   (1) Linearity          -- visual check with scatter + LOESS
        |   (2) Normal residuals   -- Shapiro-Wilk test
        |   (3) Equal variances    -- Levene's test
        |   (4) Parallel slopes    -- interaction model test
        v
STEP 5: Assumptions Checked
        |
        | Fit ANCOVA model:
        |   Memory = Voice + Practice_RT + Practice_Error
        |
        | Extract: F-statistics, p-values, partial eta-squared
        v
STEP 6: ANCOVA Results
        |
        | Compare: adjusted means vs raw means
        | Visualise: scatter plots for each covariate
        v
STEP 7: FINAL CONCLUSION
        If Practice RT or Practice Errors are significant (p < 0.05):
            --> H0 rejected: practice performance predicts memory
            --> Future models should include these as covariates
        Else:
            --> H0 retained: practice performance does NOT predict memory
```

```
--> Main experiment effects are independent of practice performance
```

10.1 Summary Table

Table 12: Table 9. Complete H5 Summary – from data to decision

What We Checked	Result
Data Source	114 log files from Sentence Memorability experiment
Practice Covariates	covariate_RT (practice speed), covariate_Error (practice errors)
Main Outcome (DV)	Corrected Memorability = Hit Rate minus False Alarm Rate
Main Group Variable (IV)	Grammatical Voice (Active or Passive)
Statistical Test	ANCOVA with Type III sums of squares and partial eta-squared
Assumption 1: Linearity	Checked visually with scatter plot and LOESS smoother
Assumption 2: Normal Residuals	Violated (Shapiro-Wilk p = 0)
Assumption 3: Equal Variance	Met (Levene p = 0.086)
Assumption 4: Parallel Slopes	Met – slopes are parallel across Voice conditions
Practice RT significant?	No (p = 0.396)
Practice Errors significant?	Yes (p = 0.001)
Overall Model significant?	Yes (p = 0.001)
Final Decision	**REJECT H0: Practice performance predicts memory scores**